

Oligopoly

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The core text is N. Gregory Mankiw, *Principles of Microeconomics*.

- **Chapter 18** — Oligopoly.

Read the chapter alongside these slides.

Where we are going

1. **Competition among the few** — what an oligopoly is.
2. **One market, three outcomes** — competition, monopoly, cartel, all from one table.
3. **The temptation to cheat** — why cartels crack; Nash equilibrium.
4. **Competing on price** — same plants, different weapon, startling outcome.
5. **Reading a payoff matrix** — game theory's basic tool, built slowly.
6. **The dilemma everywhere** — OPEC, arms races, advertising.
7. **Extensions** — repeated play and the Axelrod tournament.
8. **Antitrust** — Sherman, Clayton, and what should be illegal.

PART 1

Competition Among the Few

Oligopoly. A market structure in which only a few sellers offer similar or identical products.

Oligopoly is **competition among the few**. Real examples:

- wireless carriers — three national networks;
- large passenger aircraft — Boeing and Airbus;
- airlines on a given route — often two or three.

Each seller is large enough that its own decisions move the market price.

The condition that changes the game

A wheat farmer never asks “what will the farm next door do?” — its choice cannot move the price. A monopolist has no rival to ask about.

An oligopolist must ask. With few sellers,

- my output moves the price **you** sell at;
- your output moves the price **I** sell at.

Strategic interdependence. Each firm's best choice depends on the choices the other firms make — and every firm knows it.

Before choosing, each firm must predict its rivals' choices.

Concentration ratio. The percentage of a market's total output supplied by its four largest firms.

From the U.S. Economic Census:

- most industries: four-firm concentration **below 50%**;
- cigarettes, large aircraft, express delivery: **above 90%**.

A high concentration ratio is the statistician's flag that a market is an oligopoly — a few firms supply nearly everything.

The four market structures

structure	sellers	product	studied
perfect competition	many	identical	done
monopolistic competition	many	differentiated	done
oligopoly	few	similar or identical	here
monopoly	one	no close substitute	done

This fills the last cell of the map: **few sellers**, whose fortunes are chained together.

Two sellers, and a new toolkit

Duopoly. An oligopoly with exactly two sellers — the simplest case, carrying all the logic.

Game theory. The study of how people behave in strategic situations, where each person's best choice depends on the choices of others.

The plan: first a duopoly worked **entirely in numbers** — then the general tool.

Active Learning 1

- (a) A town's six grocery stores have market shares 30, 25, 20, 10, 10, 5 (percent). **Compute the four-firm concentration ratio.**
- (b) Another market has ten firms with 10% each. **Compute its ratio.** Which market is the oligopoly?
- (c) The two largest stores in (a) are setting the coming week's prices. **What question must each ask that a wheat farmer never asks?** One sentence. **The reason is the marks.**

Three minutes. Paper and pencil.

PART 2

One Market, Three Outcomes

Write the equation

Our running market: a valley town with exactly **two ready-mix concrete plants** — **Ridge Concrete** and **Vale Concrete**. Identical product; buyers care only about price.

$$P = 60 - Q \quad (Q \text{ yards of concrete per day, } P \text{ in dollars})$$

Each plant can produce any amount at **marginal cost \$12 per yard**, with no fixed cost (a simplification — it keeps profit = $(P - 12) \times$ yards sold).

Plug the numbers in

Q	$P = 60 - Q$	$TR = P \times Q$	$TC = 12 \times Q$	profit = $TR - TC$
0	60	$60 \times 0 = 0$	0	0
6	54	$54 \times 6 = 324$	72	252
12	48	$48 \times 12 = 576$	144	432
18	42	$42 \times 18 = 756$	216	540
24	36	$36 \times 24 = 864$	288	576
30	30	$30 \times 30 = 900$	360	540
36	24	$24 \times 36 = 864$	432	432
42	18	$18 \times 42 = 756$	504	252
48	12	$12 \times 48 = 576$	576	0
54	6	$6 \times 54 = 324$	648	-324
60	0	$0 \times 60 = 0$	720	-720

One table. Every outcome here is already sitting in it.

Reading 1: the competitive benchmark

Suppose the town's concrete came from **many** small plants, each with $MC = 12$.

- Any price above \$12 leaves profit on the table — some plant undercuts to win the business.
- Below \$12, every yard sells at a loss — no plant offers it.

So competition drives price to marginal cost:

$$P = 12, \quad Q = 60 - 12 = 48, \quad \text{profit} = 576 - 576 = 0.$$

Competitive outcome: $P = MC$, the table's zero-profit row.

Reading 2: the monopoly benchmark

Now suppose **one owner** ran both plants — a monopoly. It reads the profit column and picks the peak:

Q	18	24	30
profit	540	576	540

$$Q = 24, \quad P = 36, \quad \text{profit} = (36 - 12) \times 24 = 576.$$

Monopoly outcome: half the competitive output, three times the price.

Check it a second way

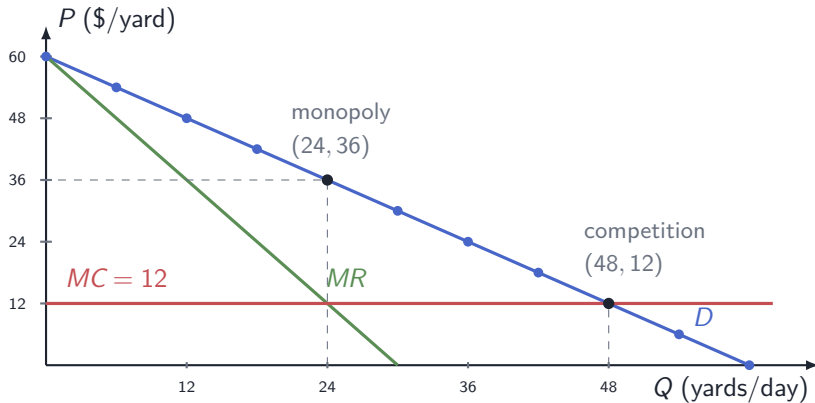
Marginal revenue from the table — each 6-yard block's $\Delta TR/\Delta Q$ — against $MC = 12$:

block	6–12	12–18	18–24	24–30	30–36
<i>MR</i> per yard	42	30	18	6	–6
vs. $MC = 12$	produce	produce	produce	stop	stop

Expanding pays through $Q = 24$ ($MR = 18 > 12$) and stops there ($MR = 6 < 12$).

Same answer by a different route: $Q = 24$. The two readings agree.

Draw the graph



Reading 3: two plants, one plan

Collusion. An agreement among firms in a market about the quantities to produce or the prices to charge.

Cartel. A group of firms acting in unison.

If Ridge and Vale collude, the market is in effect served by a **monopoly**: they maximize *joint* profit by producing the monopoly quantity, $Q = 24$.

The cartel agreement

The deal: **12 yards each**, half the monopoly quantity.

market quantity	$12 + 12 = 24$
price	$60 - 24 = 36$
each plant's profit	$(36 - 12) \times 12 = \mathbf{288}$
joint profit	$288 + 288 = 576$

Jointly, this is as good as it gets: no other agreement earns the pair more than 576.

One question hangs over the handshake: **will it hold?**

Active Learning 2

Ridge and Vale consider a *different* pact: **15 yards each** ($Q = 30$).

- (a) **Predict before computing:** will each plant earn more or less than the 288 of the 12-each deal? Why?
- (b) Compute the price, then each plant's profit **two ways**: as $TR - TC$, and as $(P - MC) \times \text{yards}$. They must agree.
- (c) Which pact should they sign? **The reasoning is the marks.**

Four minutes. Paper and pencil.

PART 3

The Temptation to Cheat

Ridge's private arithmetic

Suppose Vale honors the deal and mixes 12. If Ridge mixes q , then $Q = q + 12$, so $P = 48 - q$ and

$$\text{Ridge's profit} = (P - 12) \times q = (36 - q) q.$$

Ridge's q	12	14	16	18	20
profit $(36 - q)q$	288	308	320	324	320

Cheating pays: 18 yards earns 324, against 288 for honoring the deal.

The betrayal is mutual

Vale's desk has the same calculator. If Ridge mixes 12, Vale's best move is also 18.

And if **both** cheat to 18:

$$Q = 36, \quad P = 60 - 36 = 24, \quad \text{each: } (24 - 12) \times 18 = 216.$$

Both worse off than under the deal: $216 < 288$. And 18 is no longer anyone's best move — against a rival at 18, the best reply is 15, not 18.

The outputs keep shifting. Where does the churning stop?

The resting point

Try **16 yards each**: $Q = 32$, $P = 60 - 32 = 28$, each earns $(28 - 12) \times 16 = 256$.

Test it. Against a rival at 16, my profit is $(48 - 16 - q)q = (32 - q)q$:

my q	15	16	17
my profit	255	256	255

At 16 against 16, no deviation helps either firm. Nobody moves. The churning is over.

Nash equilibrium

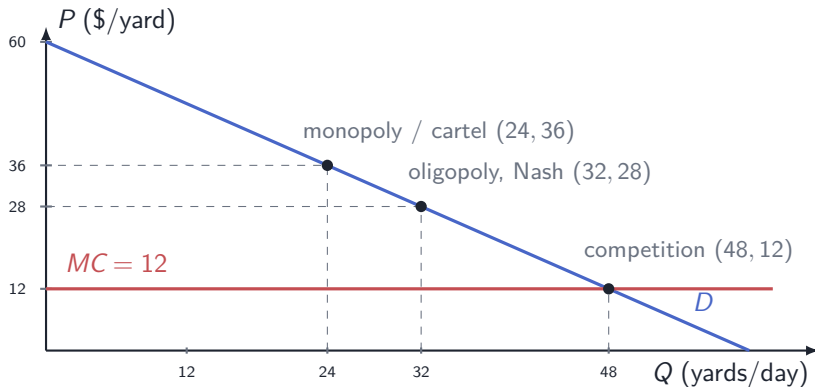
Nash equilibrium. A situation in which each economic actor, interacting with others, chooses their best strategy *given* the strategies all the other actors have chosen.

Named for John Nash (Nobel Prize, 1994). The test, applied:

candidate	the deviation test	verdict
12 & 12 (cartel)	deviating to 18 earns $324 > 288$	not an equilibrium
16 & 16	every deviation earns $\leq 255 < 256$	Nash equilibrium

An equilibrium is a pair of choices that survives both firms' private arithmetic.

Three outcomes, one market



Output: $24 < 32 < 48$. Price: $36 > 28 > 12$.

What if a third plant opened?

Run the same deviation test with **three** plants: it rests at **12 yards each** (against rivals' 24, profit $(24 - q)q$ peaks at $q = 12$: try 11 or 13 — both earn $143 < 144$).

plants	market Q	P	total profit
1 (monopoly)	24	36	576
2 (our duopoly)	32	28	512
3	36	24	432
many (competitive)	48	12	0

Each extra seller drags the market closer to the competitive outcome.

Active Learning 3

Same market: $P = 60 - Q$, $MC = 12$, two plants.

- (a) **Predict first:** at the cartel position (12 & 12), is each firm tempted to produce *more or less*? One sentence of reasoning.
- (b) **Show 12 & 12 is not a Nash equilibrium:** find a deviation for one firm that raises its profit, and compute the gain.
- (c) **Show 16 & 16 is one:** compute the deviator's profit at 15 and at 17 and compare with 256.

Five minutes. Paper and pencil — the deviation test is the marks.

PART 4

Competing on Price

Change the weapon, keep everything else

Same town, same demand $P = 60 - Q$, same $MC = 12$, same two plants. One change: each morning, each plant **posts a price** instead of choosing a quantity.

Concrete is identical, so:

- every buyer calls the **cheaper** plant;
- at equal prices, the plants **split** the market.

Start where the cartel would like: both post **\$36**. Market buys 24; each plant sells 12, earning $(36 - 12) \times 12 = 288$.

The undercut

Ridge shaves a dollar: posts **\$35** against Vale's \$36.

	Ridge (\$35)	Vale (\$36)
yards sold	$60 - 35 = 25$ (all of it)	0
profit	$(35 - 12) \times 25 = \mathbf{575}$	0

One dollar off the sign, and profit **doubles** — 575 against 288. Vale's reply: post \$34 and take it all back: $(34 - 12) \times 26 = 572$.

Any price above \$12 hands a rival the same move.

Where the undercutting stops

At $P = 12$: $Q = 48$, split 24 each, profit $(12 - 12) \times 24 = 0$.

Run the deviation test:

- undercut to \$11? Sell $Q = 49$, profit $(11 - 12) \times 49 = -49$ — a **loss**. Nobody cuts below cost.
- raise to \$13? Sell **nothing**. Profit 0 — no gain.

$P = MC = 12$ is the Nash equilibrium of the price game: **two** firms, the *competitive outcome*.

After Joseph Bertrand (1883), who first worked out price-setting rivalry.

The punchline

Same two plants, same demand, same costs — different weapon:

the duopoly competes on...	Q	P	profit each
quantities	32	28	256
prices	48	12	0

How competitive a market is depends not only on how *many* firms it has,
but on *what they compete over*.

Active Learning 4

Price competition, both plants posting **\$20**. ($P = 60 - Q$, $MC = 12$; cheaper plant takes the whole market, ties split.)

- (a) Compute each plant's profit at the tie.
- (b) Ridge undercuts to \$19: compute its new profit. Worth it?
- (c) **Predict, then verify:** at what posted price does undercutting stop paying? Show the check at that price — **why** does one more dollar of undercutting fail there?

Four minutes. Paper and pencil.

PART 5

Reading a Payoff Matrix

A game has three parts

Game. A strategic situation described by three lists: the **players**, the **strategies** each may choose, and the **payoff** each receives for every combination of choices.

- **Players** — who is choosing. (Ridge and Vale were players.)
- **Strategies** — the complete menu of actions open to each. (Mix 12 or mix 16.)
- **Payoffs** — what each player gets for *each pair* of choices. (The profits we computed.)

Specify all three, and the game is fully described — nothing else is needed.

Two prisoners

Mr. Red and **Mr. Blue** are arrested. The evidence supports only a minor charge — **1 year each** — unless one implicates the other. Questioned in **separate rooms**, each is offered the same deal:

- You confess, he stays silent: **you walk free** (0 years); he serves **3**.
- Both confess: **2 years each**.
- Both stay silent: the minor charge only — **1 year each**.

Vocabulary: staying silent is **cooperating** (with each other); confessing is **defecting**.

The payoff matrix

	Blue: cooperate	Blue: defect
Red: cooperate	(1, 1)	(3, 0)
Red: defect	(0, 3)	(2, 2)

How to read it — three rules:

- Mr. Red picks the **row**; Mr. Blue picks the **column**.
- Each cell shows **years in jail**: Red's first, Blue's second. **Fewer is better**.
- Neither picks a cell — a cell is where *two* choices land together.

Reading a strategy pair

Practice each cell — upper-left, around the grid:

- (cooperate, cooperate) $\rightarrow$ (1, 1): both silent, minor charge only.
- (cooperate, defect) $\rightarrow$ (3, 0): Red held out, Blue talked — Red serves 3, Blue walks.
- (defect, cooperate) $\rightarrow$ (0, 3): the mirror image.
- (defect, defect) $\rightarrow$ (2, 2): both talked, 2 years each.

A strategy pair names one choice per player; the matrix then hands each his payoff.

Dominant strategy

Dominant strategy. A strategy that is best for a player *no matter what* strategy the other players choose.

Find Red's, one column at a time (fewer years is better):

if Blue plays...	Red cooperates	Red defects	better
cooperate	1	0	defect
defect	3	2	defect

Whatever Blue does, Red serves fewer years by defecting: defection is dominant.

Equilibrium by elimination

The game is symmetric: the same column-by-column check makes **defect dominant** for **Blue** too.

Eliminate what no rational player chooses — each man's dominated “cooperate” row or column — and one cell survives:

$$(\text{defect}, \text{defect}) \rightarrow (2, 2).$$

Check it with the **Nash test**: at $(2, 2)$, either man who switches alone to silence goes from 2 years to **3**. No profitable deviation — the same test the duopoly passed at 16 & 16.

Prediction: both confess.

The paradox

	Blue: cooperate	Blue: defect
Red: cooperate	(1, 1) better for both	(3, 0)
Red: defect	(0, 3)	(2, 2) equilibrium

Two rational players, each choosing flawlessly, land on an outcome **both** regard as worse.

Individual rationality $\neq$ collective rationality.

The cartel is the same game

Strategies from part 3: **low** (12, the cartel share) or **high** (16). Payoffs: the profits we computed.

	Vale: low (12)	Vale: high (16)
Ridge: low (12)	(288, 288)	(240, 320)
Ridge: high (16)	(320, 240)	(256, 256)

Profits: **more** is better. High dominates ($320 > 288$; $256 > 240$) — equilibrium (256, 256), with (288, 288) unreachable.

The prisoners' matrix, with profits for years. The cartel's collapse *is* the dilemma.

Active Learning 5

Food trucks **Grill** and **Cart**: **keep** prices or **discount**. Weekly profits:

	Cart: keep	Cart: discount
Grill: keep	(500, 500)	(300, 650)
Grill: discount	(650, 300)	(400, 400)

- (a) Find each truck's dominant strategy — **show both comparisons** per truck.
- (b) Name the equilibrium, **verify it** with the deviation test, and say **why** this is (or is not) a prisoners' dilemma.

Four minutes. Paper and pencil.

PART 6

The Dilemma Everywhere

The fingerprint of a dilemma

A situation is a prisoners' dilemma whenever **both** hold:

1. **Cooperation is jointly best** — both players prefer (cooperate, cooperate) to (defect, defect);
2. **defection is individually dominant** — each does better defecting *whatever* the other chooses.

Whenever you can check both boxes, you already know the ending:

cooperation collapses — without communication, contracts, or repetition.

A cartel of countries: OPEC

The Organization of the Petroleum Exporting Countries — founded 1960; members include Saudi Arabia, Iran, Iraq, Kuwait, Venezuela, Nigeria, and the UAE.

- The aim: **restrict members' oil output** — shift the world market down its demand curve, as Ridge and Vale did with 12 yards each.
- Each member's quota is its “12”: and at the cartel's high price, **every member is tempted to pump beyond it.**

Same matrix, with countries for players and barrels for yards.

An arms race

Cold War superpowers, each choosing: **arm** or **disarm**.

	USA: arm	USA: disarm
USSR: arm	at risk, at risk	safe and dominant, at risk and weak
USSR: disarm	at risk and weak, safe and dominant	safe, safe

Whatever the rival does, arming is safer than being the only one disarmed: **arming is dominant** — both arm, both worse off than mutual disarmament.

An advertising war

Two rival soda makers, each choosing an ad budget.

- Rival ads largely **cancel**: each campaign mostly defends share against the other's campaign.
- If the rival advertises heavily, you must too — or lose share. If the rival does not, heavy ads **steal** share. Advertising is **dominant**.

Equilibrium: both spend heavily, market shares end where they began — **both profits lower by the cost of the campaigns**.

Check the fingerprint: jointly best to disarm, individually best to arm. The same game.

Is defection always bad?

The equilibrium disappoints the *players*. Society is a different judge:

game	players lose...	society...
cartel breakdown	profits ($256 < 288$)	gains: Q 32 not 24, P 28 not 36
arms race	security spending	loses: expense, danger
ad war	profits	largely loses: resources cancel out

When cartels play the dilemma, their failure is the *consumer's* win — the invisible hand needs the betrayal.

The lesson of the dilemma

In oligopoly there is a permanent tension between **cooperation** and **self-interest**.

- Self-interest pulls output **above** the cartel's, price **below** monopoly's — but not all the way to the competitive outcome.
- That is why oligopoly **lives between the poles**: $24 < 32 < 48$.

So far every game was played **once**. Real rivals meet again tomorrow — and that changes what is rational.

Active Learning 6

Build the matrix yourself. Two streaming services, **Redwood** and **Bluebird**, each choose a **small** or **big** ad campaign.

- With small campaigns both earn **100**.
 - A big campaign costs **20** and pulls **30** of profit from the rival — but two big campaigns cancel: nothing is pulled either way.
- (a) Draw the 2×2 payoff matrix (Redwood's payoff first, in every cell).
- (b) Find the dominant strategies and the equilibrium.
- (c) Check the fingerprint: is it a prisoners' dilemma? One sentence why.

Five minutes. Paper and pencil — constructing the matrix *is* the exercise.

PART 7

Extensions: Repeated Play and the Tournament

Once is not forever

Every game so far was played **once**. But Ridge and Vale post prices *every morning*; OPEC meets *every year*.

Repeated game. The same game, played by the same players, round after round — with each round's choices visible before the next.

What repetition adds: **today's defection can be punished tomorrow**. A strategy can now respond to the rival's past.

A threat with teeth: the grim strategy

Vale announces: *"I mix 12 as long as you always have. The first day you exceed it, I mix 16 forever."*

Ridge's arithmetic, per round (from parts 3 and 5):

- cheat today: **gain** $324 - 288 = 36$ — once;
- punished tomorrow onward: 256 instead of 288 — **lose 32 every round, forever.**

If the plants expect to meet even two more times, cheating is a losing move:

$$36 < 32 + 32.$$

The trouble with grim

Grim works — but it is **unforgiving**:

- One defection — even a **mistake**, a misread, a bad day — and cooperation is dead *forever*.
- Both players then grind out 256 a round, with 288 permanently out of reach.

Is there a strategy that punishes cheating — but lets cooperation heal?

That question was settled not by a proof, but by a **tournament**.

Axelrod's tournament

Robert Axelrod (*The Evolution of Cooperation*, 1984) invited game theorists to submit **computer programs** to play repeated prisoners' dilemma:

- every program played **every other program**, round after round;
- the winner: the **highest total payoff** across all opponents;
- entries came from psychology, economics, mathematics, political science — some enormously intricate.

The winner — submitted by Anatol Rapoport — was the **simplest entry in the field**:

1. Cooperate in the first round.
2. Then do whatever the rival did last round.

Punishes a defection **immediately** — once — and returns to cooperation the moment the rival does.

It won again when the tournament was rerun — against a field that knew it had won the first.

Why tit-for-tat wins

Axelrod's autopsy found four properties:

- **Nice** — never defects first: earns full cooperation from the willing.
- **Retaliatory** — answers defection at once: cannot be farmed by the ruthless.
- **Forgiving** — punishes one round, not forever: mistakes heal.
- **Clear** — rivals learn its rule fast, and see that cooperation pays against it.

Cooperation can emerge among pure egoists — no trust, no contract, no referee —
given a future.

What repetition means for oligopoly

Firms in an oligopoly meet **every day** — so they can sustain collusion *without ever speaking*:

- **Tacit collusion** — match rivals' prices, punish undercutting, return when they return: tit-for-tat with price lists.
- No meeting, no memo, no agreement to prosecute — yet the cartel outcome can hold.

Repetition is why oligopoly worries the law even when no one ever shakes hands.

Active Learning 7

Ridge and Vale will interact for exactly **4 rounds**, both know it, and Vale plays **grim** (12 until Ridge ever cheats; 16 forever after). Per round: cooperate $\rightarrow$ 288; cheat against 12 $\rightarrow$ 324; rounds after a cheat $\rightarrow$ 256.

- (a) Ridge cooperates all 4 rounds: **total profit?**
- (b) Ridge cheats in round 1 (then earns 256 in each later round): **total?**
- (c) Ridge cooperates in rounds 1–3 and cheats in round **4**: total? Which of the three plans wins — and **what does that say about games with a known last round?**

Five minutes. Paper and pencil. Part (c) is the thinking item.

PART 8

Antitrust

The oldest warning in economics

Adam Smith, *The Wealth of Nations* (1776):

“People of the same trade seldom meet together, even for merriment and diversion, but the conversation ends in a conspiracy against the public, or in some contrivance to raise prices.”

These matrices say Smith saw it exactly: cooperation *against consumers* is where sellers' self-interest points — **policy's job is to keep that cooperation hard.**

The Sherman Antitrust Act, 1890

Section 1 — against *agreements*:

“Every contract, combination in the form of trust or otherwise, or conspiracy, in restraint of trade or commerce among the several States... is declared to be illegal.”

Section 2 — against *monopolizing*:

“Every person who shall monopolize, or attempt to monopolize, or combine or conspire... to monopolize... shall be deemed guilty of a felony.”

Before 1890 a cartel contract was merely unenforceable. Sherman made it a **crime**.

The Clayton Act, 1914

Sherman's enforcer was the government alone. Clayton **armed the victims**:

- **Treble damages** (Section 4): a private party injured by a violation can sue for **three times** its losses.
- **Injunctive relief** (Section 16): courts can order conduct stopped — a power the Supreme Court has read to include forcing the **divestiture** of assets.

Triple damages turn every overcharged customer into a potential enforcer.

The clear case: price fixing

On one thing courts and economists agree completely: **naked price fixing** — rivals agreeing on price or output — is **per se illegal**.

- *Per se*: no defense heard. Not “the price was reasonable,” not “the industry was struggling.” **The agreement itself is the offense.**
- Why so strict? The matrix: price fixing has **no redeeming output** — it moves the market from 32 yards at \$28 toward 24 at \$36 and produces nothing.

Real prosecutions are routine — and executives serve real prison time.

Controversy 1: resale price maintenance

A manufacturer requires retailers to charge **no less than a set price** for its product.

- **The case against:** it looks like price fixing — a floor under retail prices, competition suppressed.
- **The economists' doubt:** the manufacturer *gains nothing* from fat retail margins — it may instead be buying **service**: showrooms, trained staff, demonstration — things discounters would free-ride on and kill.

Same act, two readings: suppressing competition — or purchasing retail effort.

Controversy 2: predatory pricing

A dominant firm prices **below cost** to drive a rival out — planning to recoup with high prices afterward.

- **The case against:** it is monopolizing — Sherman Section 2.
- **The economists' doubt:** the predator outsells everyone *at the loss-making price* — it bleeds fastest; and the recoupment phase's high prices **invite re-entry**, unraveling the plan.

The hard problem: a predatory price cut and a competitive price cut **look identical to the customer.**

Controversy 3: tying and bundling

Selling product A **only together with** product B — two movies licensed as a block; an operating system shipped with its maker's browser.

- **The case against:** leverage — market power in A is used to *conquer* the market for B.
- **The economists' doubt:** buyers have **one** willingness to pay for the pair — power in A can be charged for in A's price; tying does not *manufacture* extra power. The bundle may simply be **cheaper to sell**.

Still genuinely unsettled — the centerpiece of *U.S. v. Microsoft* (2001).

Active Learning 8

Three cases. For each: **clearly illegal, or genuinely controversial?** Then **one sentence of economics** for your verdict — the sentence is the marks.

- (a) Two concrete plants exchange emails agreeing both will charge \$36 per yard.
- (b) After a new plant enters, the incumbent cuts its price below cost until the entrant leaves.
- (c) A power-tool maker tells retailers they may not sell its drills below \$50.

Four minutes. Paper and pencil.

Oligopoly is a prisoners' dilemma played for money.

Self-interest keeps pulling rivals from the cartel toward competition —
repetition pulls them back —
and the law works to keep the temptation winning.